

Breaking Barriers: The Impact of High-Speed Rail Opening on Foreign-Invested Enterprise Establishment via a Regional Market Environment

Appendix

Appendix Table 1: Placebo Test Results of HSR Openings

	(1)	(2)	(3)
	1 year	2 years	3 years
<i>hsr</i>	0.0185 (0.0332)	-0.00513 (0.0135)	0.0108 (0.0125)
Control	Yes	Yes	Yes
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	2,841	2,841	2,841
Pseudo R ²	0.9383	0.9383	0.9383

Note: Appendix Table 1 shows the placebo test results of HSR openings. Several hypothetical shocks are built to detect if there is any significant effect of HSR one year, two years, and three years before its openings, respectively.

Appendix Table 2: Instrumental Variable Regression Results

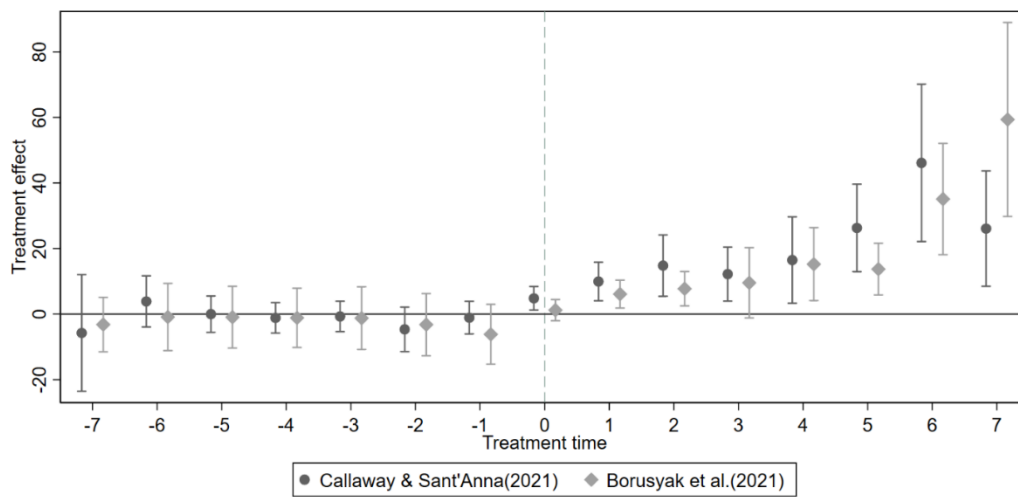
	(1)	(2)	(3)	(4)
	Ming dynasty postal stations		Relief degree of land	
	First stage	Second stage	First stage	Second stage
<i>IVI</i>	0.00310*** (0.000484)			
<i>IV2</i>			-0.00495*** (0.00183)	
<i>hsr</i>		4.064** (1.664)		1.193** (0.507)
Control	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
First stage F	42.63		29.17	
Observations	2,841	2,841	2,851	2,841
R ² /Pseudo R ²	0.384	0.9385	0.352	0.9384

Note: Instrumental variable *IVI* is a binary indicator denoting whether a city hosted a Ming-Dynasty postal station. Since this variable is time-invariant, we interact it with the city's lagged annual average relative humidity. Instrumental variable *IV2* is the city's relief degree of land. Because the relief degree of land is also time-invariant, we interact it with the lagged national ratio of HSR to total railway operating mileage. *, **, and *** represent significance at 10%, 5%, and 1%, respectively.

Appendix Table 3: The effect of HSR openings on other cities' FIEs establishment in same province

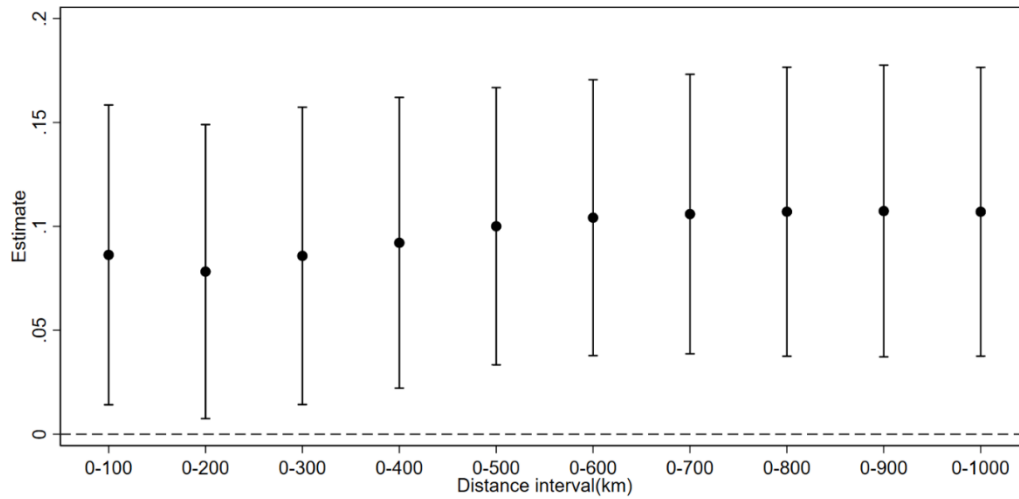
	(1)
<i>hsr</i>	0.0354 (0.0279)
Control	Yes
City FE	Yes
Year FE	Yes
N	2,806
Pseudo R ²	0.9511

Note: The dependent variable is the number of FIEs established in all other cities within the same province, excluding the city itself. We use PPML estimation and robust standard errors clustered at the city level.



Appendix Figure 1: Parallel Trend Test Results with Heterogeneous Treatment Effects

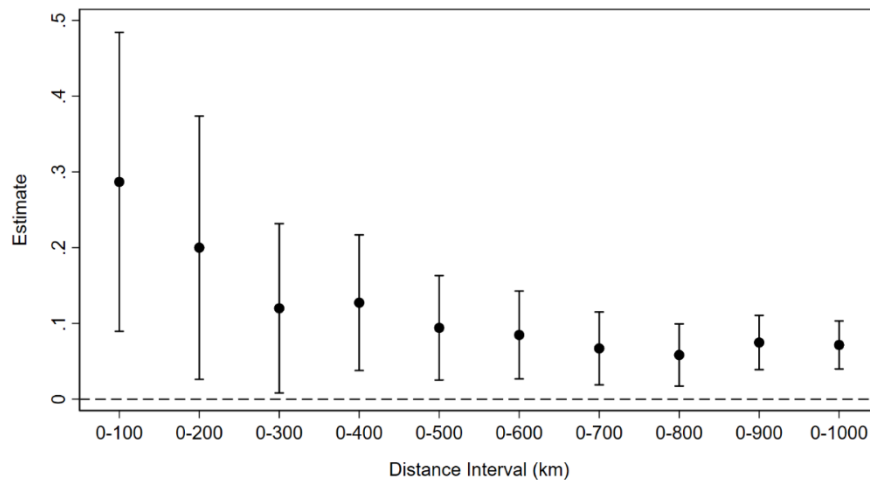
Note: Appendix figure 1 shows the parallel trend test results using the methods provided by Callaway & Sant'Anna (2021) and Borusyak et al. (2024).



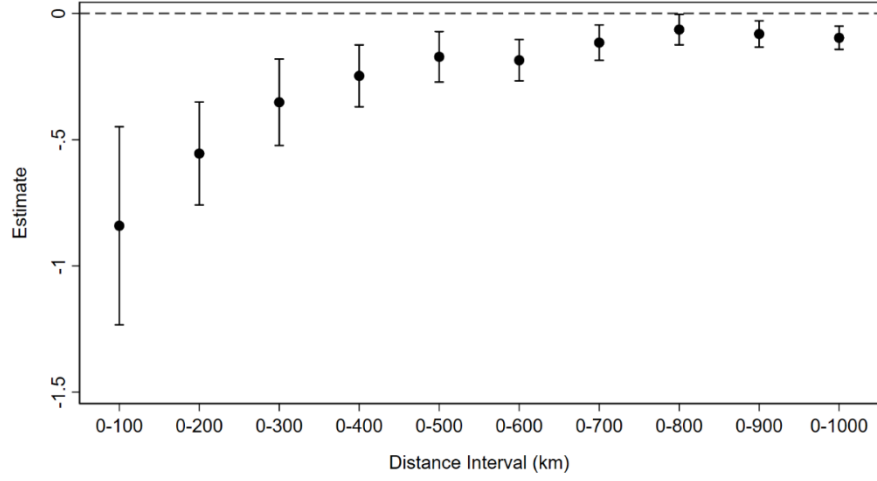
Appendix Figure 2: Controlling for the Number of HSR Stations in Surrounding Cities

Notes: Appendix figure 2 shows the results of controlling for the number of HSR stations in surrounding cities.

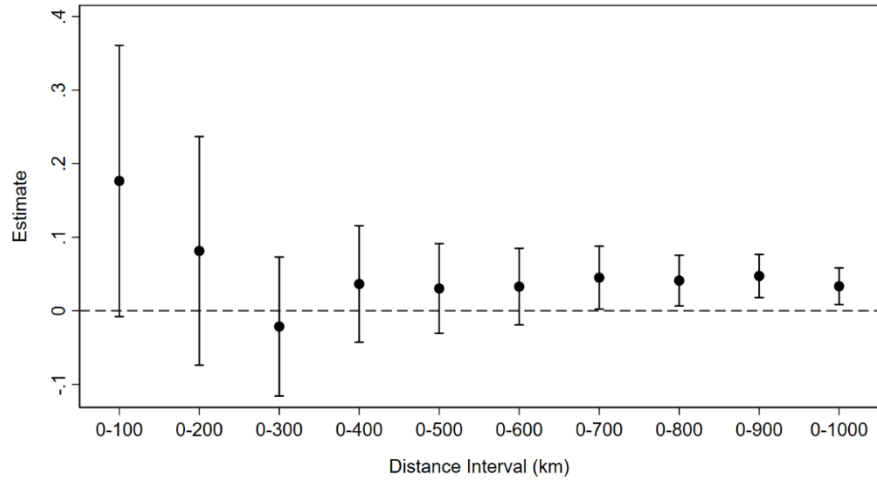
Distance unit: kilometers (km).



Appendix Figure 3: The effect of HSR on FIE establishment in neighboring HSR cities



Appendix Figure 4: The effect of HSR on FIE establishment in neighboring non-HSR cities



Appendix Figure 5: The effect of HSR on FIE establishment in neighboring HSR cities (controlling for the scale of FIE establishments in non-HSR cities)

References

- [1] Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event-study designs: robust and efficient estimation. *Review of Economic Studies*, 91(6), 3253-3285.
- [2] Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200-230.